



DAY 3 • SESSION 2

The Flight Controller, Then Wire It

Robotics, Drones & Autonomous Systems

KAUST Gifted Student Program · Summer Enrichment 2026

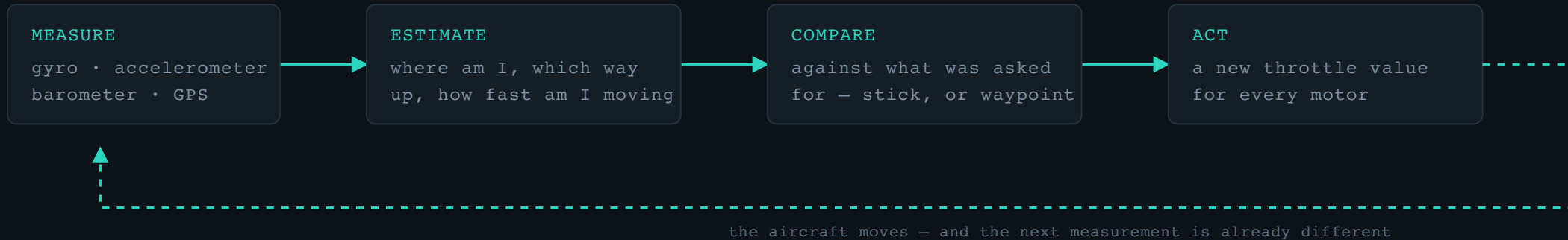
~50 minutes · FC intro + wiring + hands-on

A NEW PROJECT

You have designed an aircraft. Now meet the computer that flies it.

Two days of paper produced a rotor count, a propeller, a battery and a mass. None of it leaves the ground without the board on the next slide.

What it does, four hundred times a second



Round that loop **hundreds of times every second**, forever, while the vehicle is powered.

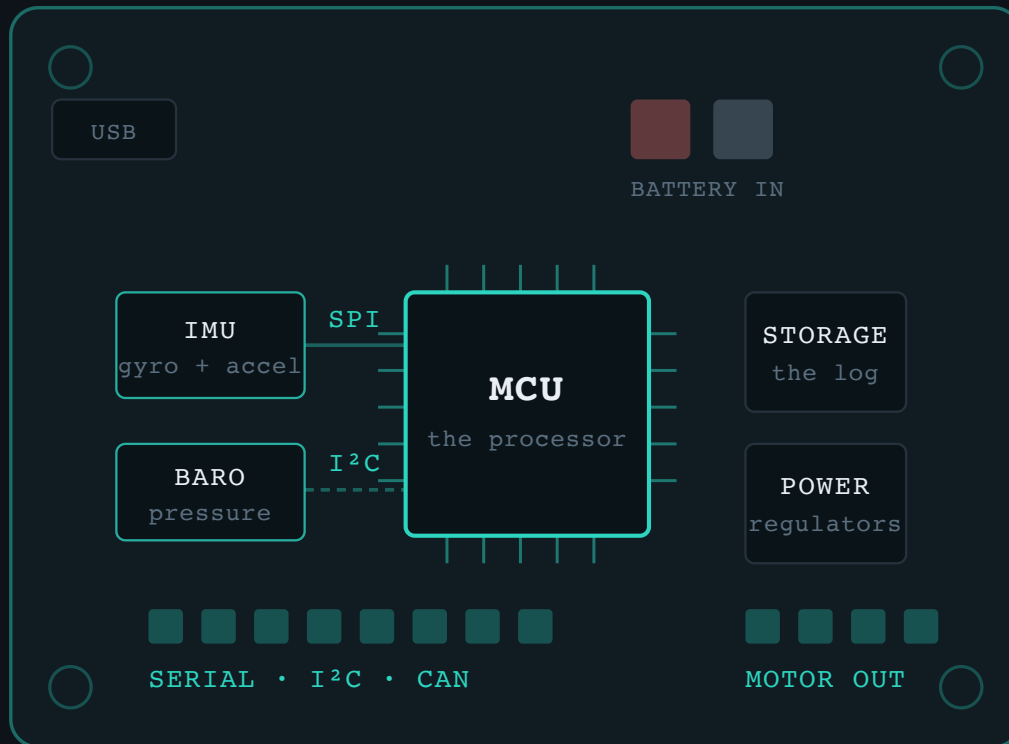
This is why a flight controller is a **microcontroller** and not a laptop. It is not doing hard arithmetic — it is doing easy arithmetic **on time, every time**.

PART TWO

What is on the board

Every flight controller made — a fifty-dollar one and a five-hundred-dollar one — has the same five things on it. Only the count and the quality change.

Five things, on every board ever made



FIXED AT THE FACTORY

Processor, inertial sensor, barometer, storage and the power regulators are **soldered down and wired to each other by traces you cannot reach**. Whoever designed the board made those decisions. You never change them.

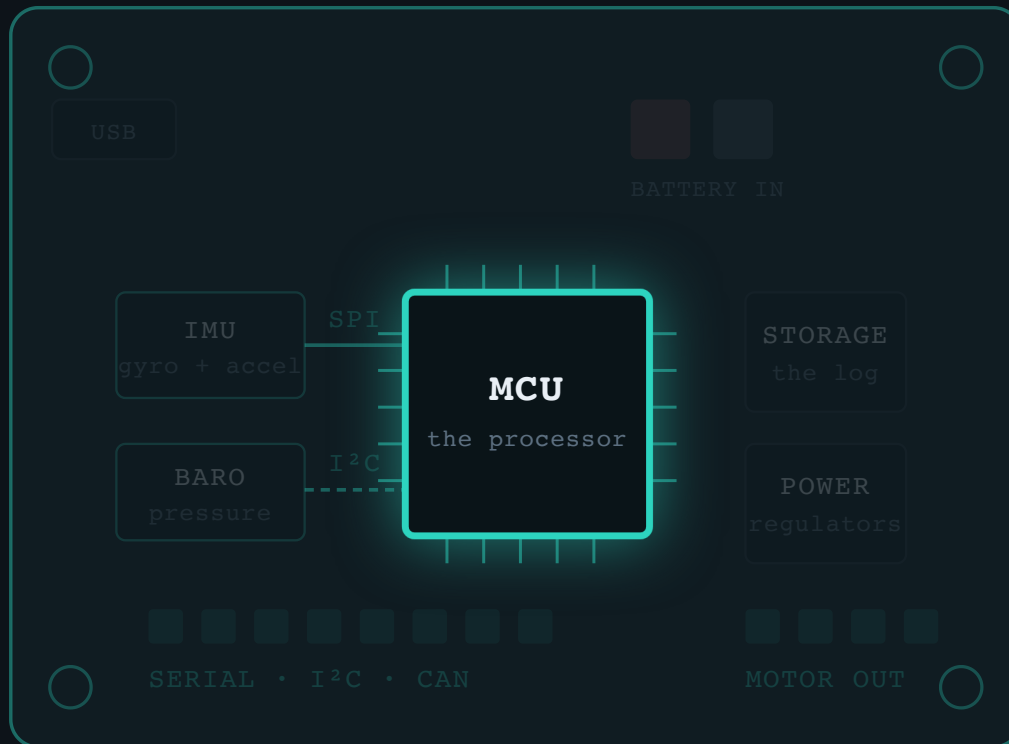
DECIDED BY YOU

The connections round the edge — serial ports, I²C, CAN, the motor outputs and the battery input. **Every one is a choice you make when you build the aircraft**, and it is the rest of this session.

THE WHOLE DESIGN, IN ONE SENTENCE

A small computer, two sensors it can trust, somewhere to write what happened, and a set of wires to the outside world.

The processor



ONE CHIP, RUNNING ONE PROGRAM

A **microcontroller**: the processor, its memory and all its input and output hardware on a single chip. It boots in milliseconds and runs your firmware from the moment power arrives until it is taken away.

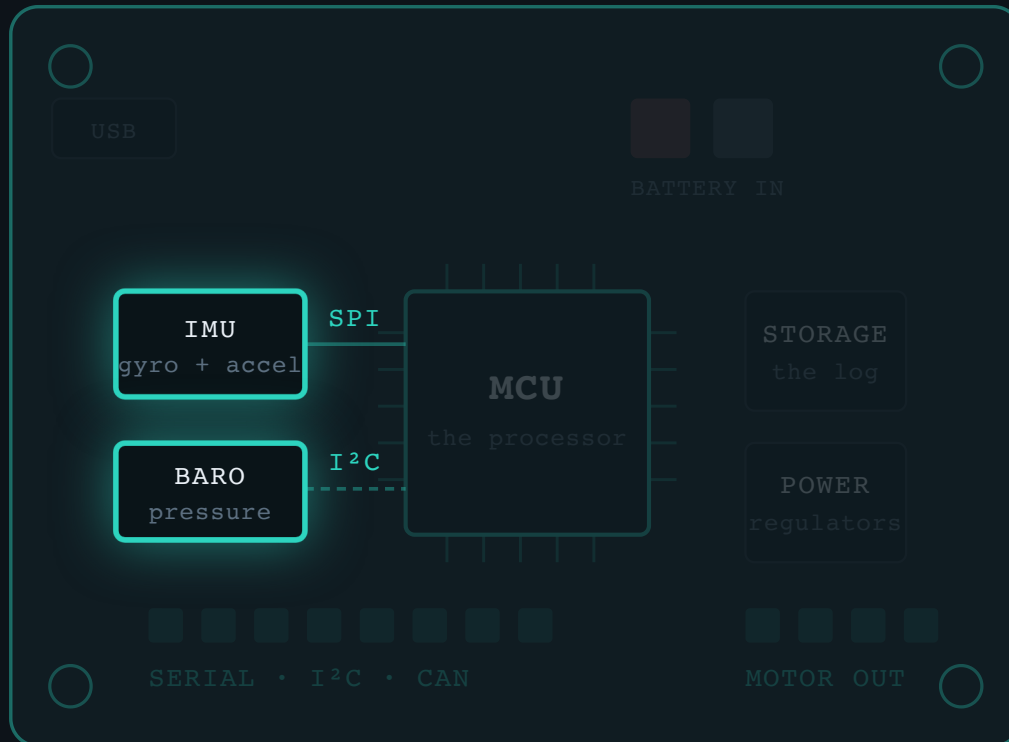
WHAT MATTERS ABOUT IT

Not how fast it is. **How reliably on time it is.** If the sensor reading due at 2.5 milliseconds arrives at 40 instead, the aircraft has already tipped over.

Modern boards run an Arm Cortex-M7 at a few hundred megahertz, with a couple of megabytes of flash holding the firmware. Your Pixhawk 6C is one of them.

Everything else on the board exists to give this chip something to read, or somewhere to send the answer.

The two sensors that never leave the board



IMU • INERTIAL MEASUREMENT UNIT

A **gyroscope** measuring how fast the aircraft is rotating about each axis, and an **accelerometer** measuring acceleration along each — including the one gravity causes.

The gyro is what actually flies the aircraft, read **thousands of times a second**. Neither instrument is enough alone: integrate the gyro and it drifts, trust the accelerometer and every vibration becomes a lie about which way is up.

BAROMETER • AIR PRESSURE

Pressure falls predictably as you climb, so this is your **altitude** — good to a few centimetres, and read only tens of times a second because altitude does not change fast.

You will not wire either of these. They are already wired, and that is the point — **everything inside the box was decided for you**. What is left is the outside of the box, and that is entirely yours.

PART THREE

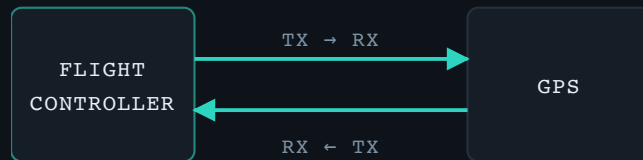
Connecting things to it

Three ways to attach a peripheral

Anything you add to the aircraft — a GPS, a radio, a rangefinder, a compass — arrives on one of these three. Which one is decided by the device, not by you.

SERIAL • UART

one device, one port

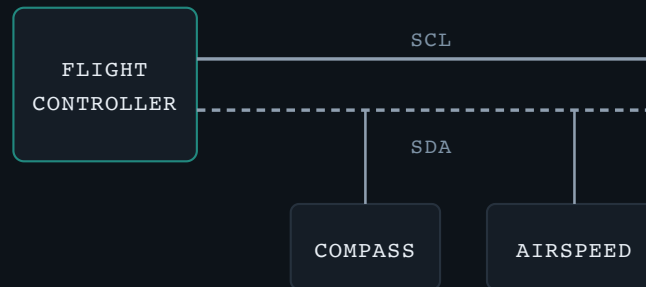


Two wires and a ground. No addresses, so no one else can share the line.

GPS • telemetry radio • companion computer • rangefinder

I²C

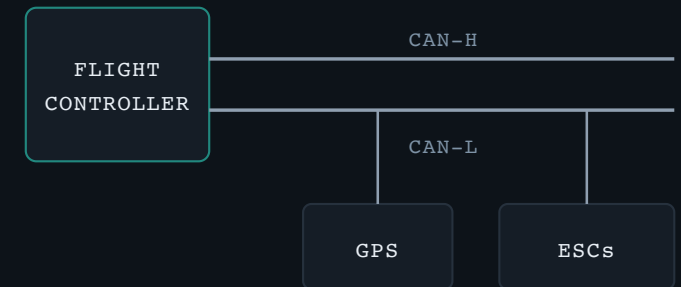
many devices, two wires



Every device has an address, so they queue. Slow, and short cables only.

CAN

many devices, long cables



The two wires carry the same signal inverted, so noise cancels out.

A serial port is a private conversation between exactly two things — **so the number of serial ports on your board is a hard limit on how many devices you can add.** I²C and CAN exist because of that limit: when you run out, the question becomes whether the next device can live on a shared bus instead.

The whole aircraft, as five cables



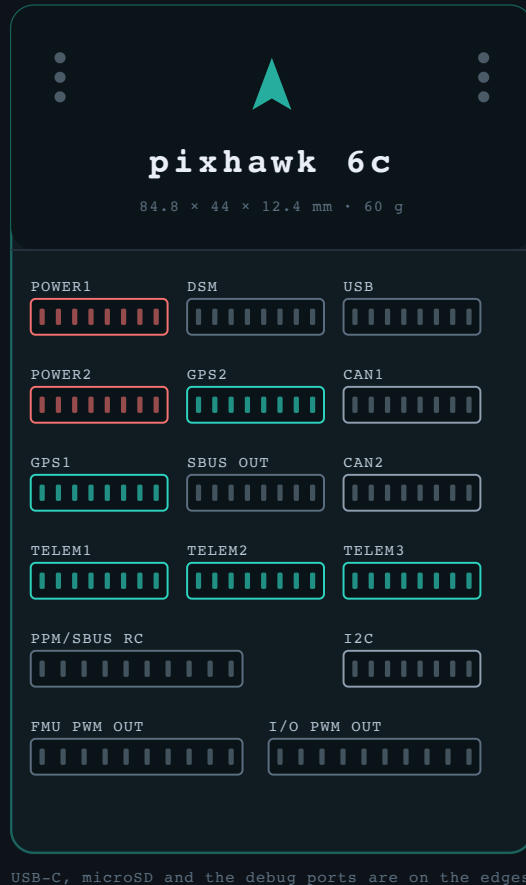
Five cables, and the aircraft is wired. **Signal and power are separate stories** — the flight controller tells the ESCs what to do, but the hundred amps that spin the motors come straight off the battery and never touch the board.

PART FOUR

Your board: the Pixhawk 6C

Everything from Part Two, in a case, with a keyed connector for every one of them.

You just wired it with five cables. It has sixteen connectors.



COLOUR TELLS YOU WHAT IT CARRIES

GPS1 · GPS2 · TELEM1 · TELEM2 · TELEM3 are the five **serial ports**. POWER1 · POWER2 take the battery through a power module. CAN1 · CAN2 · I2C are the shared buses. The rest are radio control in, and motor outputs.

EVERY CONNECTOR IS KEYED

The plug only fits one way, and the pin order is **the same on every Pixhawk-standard board ever made**. The cable in the box is already correct. That, and the second processor, is most of what you paid for.

ONE RAIL FEEDS NEARLY ALL OF THEM

Every port with a 5 V pin draws from the same regulator, and Holybro state the limit: **1.5 A for TELEM1 on its own, and 1.5 A shared by everything else combined**.

A rail that runs out in flight reboots the flight controller — a hardware failure that looks exactly like a software one.

Five serial ports — and you have used two

The GPS is on **GPS1** and the telemetry radio is on **TELEM1**. That leaves three, and the label on the case does not restrict what you put there.



USB-C, microSD and the debug ports are on the edges

STILL
FREE

WHAT YOU COULD PUT ON IT

TELEM2

Companion computer — a small Linux board doing vision or planning, asking the autopilot politely

TELEM3

Rangefinder — precise height above ground for landing, terrain following or low inspection

GPS2

Second GPS, or anything else serial — an optical flow sensor, an ADS-B receiver, a camera trigger

THE LABEL IS NOT THE LAW

A rangefinder plugged into a port marked **GPS2** works perfectly well. The case was printed at the factory; **what the port actually does is set in the firmware**, and that is the last slide of this part.

Three spare ports is a real number with a real consequence. **If your mission needs four more serial devices, you already have a problem** — and this is the moment to find it, not on Thursday.

Inside one connector: TELEM1

Six pins, JST-GH 1.25 mm. Your telemetry radio goes here — and from tomorrow, every message between the aircraft and the laptop travels down it.

PIN	SIGNAL	LEVEL	WHAT IT IS FOR
1	VCC	+5 V	Powers the radio — up to 1.5 A, its own budget
2	TX	+3.3 V	The flight controller talking
3	RX	+3.3 V	The flight controller listening
4	CTS	+3.3 V	“I am ready for more” — optional
5	RTS	+3.3 V	“Stop sending for a moment” — optional
6	GND	0 V	The shared zero

THE THREE THAT ALWAYS MATTER

TX, RX and GND. Every serial device on this aircraft needs those three and nothing more. Pins 4 and 5 are flow control, and most radios do not use them.

TX GOES TO RX

The flight controller's **talking** wire connects to the device's **listening** wire, and the other way round. On a keyed connector this is done for you. On anything you make yourself it is not — and TX wired to TX looks exactly like a dead device.

SAME SIX PINS, DIFFERENT PORT

TELEM2 is identical. So is TELEM3, minus the flow control. Learn one connector and you have learned all three.

And one connector carrying five different things

GPS1, ten pins. The unit on your GPS mast is not only a GPS, so the cable to it is not only a serial port.

PIN	SIGNAL	BELONGS TO
1	VCC • +5 V	power for the whole mast
2	TX	the GPS — a serial port, exactly like TELEM1
3	RX	
4	SCL	the compass — on I ² C, the shared bus from Part Three
5	SDA	
6	SAFETY_SWITCH	the safety button and its light
7	SAFETY_LED	
8	3V3	power for the button
9	BUZZER-	the buzzer
10	GND	the shared zero

WHY THE COMPASS IS UP A MAST

A compass measures the Earth's magnetic field, which is very weak. Your motors pull tens of amps through wires a few centimetres from the flight controller, and every one of those wires makes a magnetic field of its own — **changing with throttle**.

So the compass moves as far from the power wiring as the airframe allows. That means up a mast, and once a cable is going up there anyway, the GPS goes with it.

ONE CABLE, ONE POWER BUDGET

All five of those things draw from the same 5 V pin. The mast is the single hungriest peripheral on the aircraft — and it is on the shared allowance, not TELEM1's.

The connector is hardware. The number is a setting.

WRITTEN ON THE CASE	ARDUPILOT CALLS IT	NORMALLY CARRIES
USB	SERIAL0	your laptop, on the bench
TELEM1	SERIAL1	telemetry radio
TELEM2	SERIAL2	second radio, or a companion computer
GPS1	SERIAL3	GPS
GPS2	SERIAL4	second GPS, or a rangefinder
TELEM3	SERIAL5	spare

PLUGGING IT IN IS HALF THE JOB

The firmware has no idea what you connected. Put a rangefinder on GPS2 and nothing happens until you tell it — by setting SERIAL4_PROTOCOL to the rangefinder protocol and SERIAL4_BAUD to the right speed.

Two settings, and the port becomes the thing you plugged into it.

THIS AFTERNOON

Those settings are called **parameters**, there are about twelve hundred of them, and they are the entire reason one piece of firmware flies your quadcopter and a twenty-five kilogram hexacopter without a line being recompiled.

A port you have wired and not configured behaves exactly like a port you forgot to wire. **When a device “does not work”, this is the first place to look** — long before you suspect the device.

YOUR KIT

Six components, five cables

Everything on this table connects to one board. The order you wire them matters less than knowing what each one is before you start.

What is in front of you

COMPONENT	WHAT IT DOES	CONNECTS TO	CONNECTOR
Pixhawk 6C	Flight controller — the board you just met	everything below	JST-GH 1.25
PM02 V3	Power module — steps battery voltage down to 5 V, measures volts and amps	POWER1	JST-GH 6-pin
Micro M10 GPS	GPS + compass + safety switch + buzzer — all on one mast	GPS2	JST-GH 6-pin
RP4TD receiver	RC receiver — ELRS 2.4 GHz, takes commands from the pilot's radio	RC IN	JST-GH 4-pin
SiK Radio V3	Telemetry radio — 433 MHz, links the aircraft to the ground station laptop	TELEM1	JST-GH 6-pin
4S 5500 mAh LiPo	Battery — 14.8 V nominal, XT60 connector, the only source of power	PM02 input	XT60

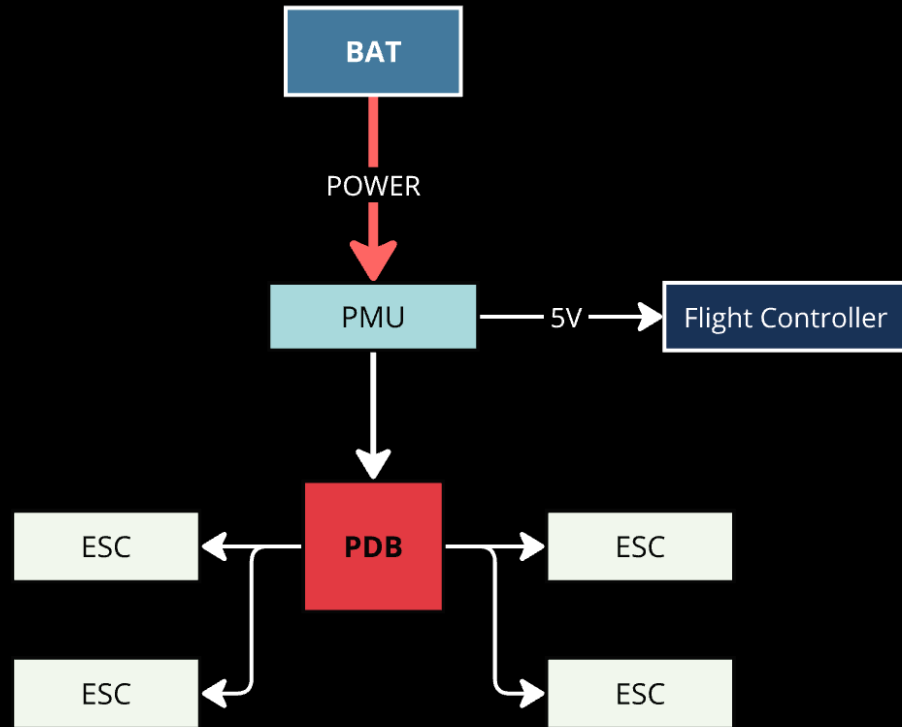
Every connector is keyed — the plug only fits one way, and the pinout is printed on the board. **If it does not click, it is not the right port.**

C A B L E O N E

The power chain

Nothing works until the board has power. The battery does not plug into the flight controller — it plugs into a module that protects the board and measures what the battery is doing.

Battery → PM02 → Pixhawk



WHY A POWER MODULE

A 4S LiPo runs at **14.8–16.8 V**. The Pixhawk needs **5 V**. The PM02 does three things: regulates 5 V for the board, measures battery voltage, and measures current draw — so the autopilot always knows how much energy is left.

XT60 — THE BIG CONNECTOR

The yellow connector between the battery and the PM02. Rated for **60 A continuous**. It is polarised — the flat pins only go one way. **Never force it.**

THE OTHER SIDE OF THE PM02

Two outputs: a **JST-GH 6-pin** cable to POWER1 on the Pixhawk (5 V + sense lines), and **XT60 pass-through** to the power distribution board or ESCs — the high-current path that spins the motors.

The motor current never enters the flight controller. **Signal and power are separate stories** — the PM02 is the boundary between them.

GPS mast — one cable, five things inside it

The Holybro Micro M10 plugs into **GPS2** with a 6-pin JST-GH cable.

WHAT IS ON THE MAST

GPS	u-blox M10 — position, velocity, time
Compass	IST8310 magnetometer — heading, on I ² C
Safety switch	Press-and-hold to arm — a physical gate
Buzzer	Audible arm/disarm, GPS lock, low battery
LED	Safety status — solid when armed

WHY IT IS ON A MAST

The compass measures the Earth's magnetic field, which is **very weak**. Your motors pull tens of amps through wires centimetres away — every one of those makes a magnetic field that changes with throttle. The mast puts distance between the compass and that noise.

MOUNTING

The mast stands on top of the frame, as far from the power wiring as the airframe allows. The cable routes down through the frame to the GPS2 connector on the Pixhawk.

ONE CABLE, ONE POWER BUDGET

All five things on the mast draw from the same 5 V pin on GPS2. The mast is the **single hungriest peripheral** on the shared 1.5 A rail — leave headroom for it.

Plug it in, mount it high, route the cable away from power wires. **If the compass calibration fails later, this cable routing is the first thing to check.**

RC receiver — the pilot's link

The RadioMaster RP4TD talks to the Pocket Crush transmitter over **ELRS 2.4 GHz**. It connects to the Pixhawk's **RC IN** port.

WHAT ELRS IS

ExpressLRS — an open-source radio protocol. Low latency, long range, and the receiver and transmitter update their own firmware over Wi-Fi. The RP4TD is a **diversity receiver**: two antennas, and it picks whichever signal is stronger on every packet.

THE PROTOCOL ON THE WIRE

The receiver speaks **CRSF** (Crossfire) to the Pixhawk — a single-wire bidirectional serial protocol. The cable in the box is a 4-pin JST-GH: **VCC**, **TX**, **RX**, **GND**.

ANTENNA PLACEMENT

Mount the two antennas in a **V or T** — at roughly 90° to each other. Keep them **away from carbon fibre** (it shields RF) and away from the ESCs and power wires (they radiate noise).

An antenna trapped under a carbon plate or zip-tied flat against the frame is the most common cause of a failsafe on the first flight.

BINDING

The receiver and transmitter must be **bound** — paired to each other. The RP4TD ships unbound. Binding is a firmware step done after the hardware is wired, not now.

Wire it, mount the antennas in a V, and leave the binding for the firmware session. **If the receiver LED blinks fast, it is looking for a transmitter — that is correct for now.**

Telemetry radio — the ground station link

The Holybro SiK V3 is a pair: one on the aircraft plugged into **TELEM1**, one on the laptop via USB. **433 MHz, 100 mW.**

WHAT IT CARRIES

MAVLink — every message between the aircraft and the ground control station. Attitude, position, battery voltage, warnings, waypoints, parameter changes, arm commands. Everything you see on the GCS screen arrived through this radio.

GROUND SIDE: USB

The second radio plugs into the laptop's USB port. The GCS sees it as a serial port. **No driver needed** on Windows, macOS or Linux — it enumerates as a standard FTDI device.

AIRSIDE: TELEM1

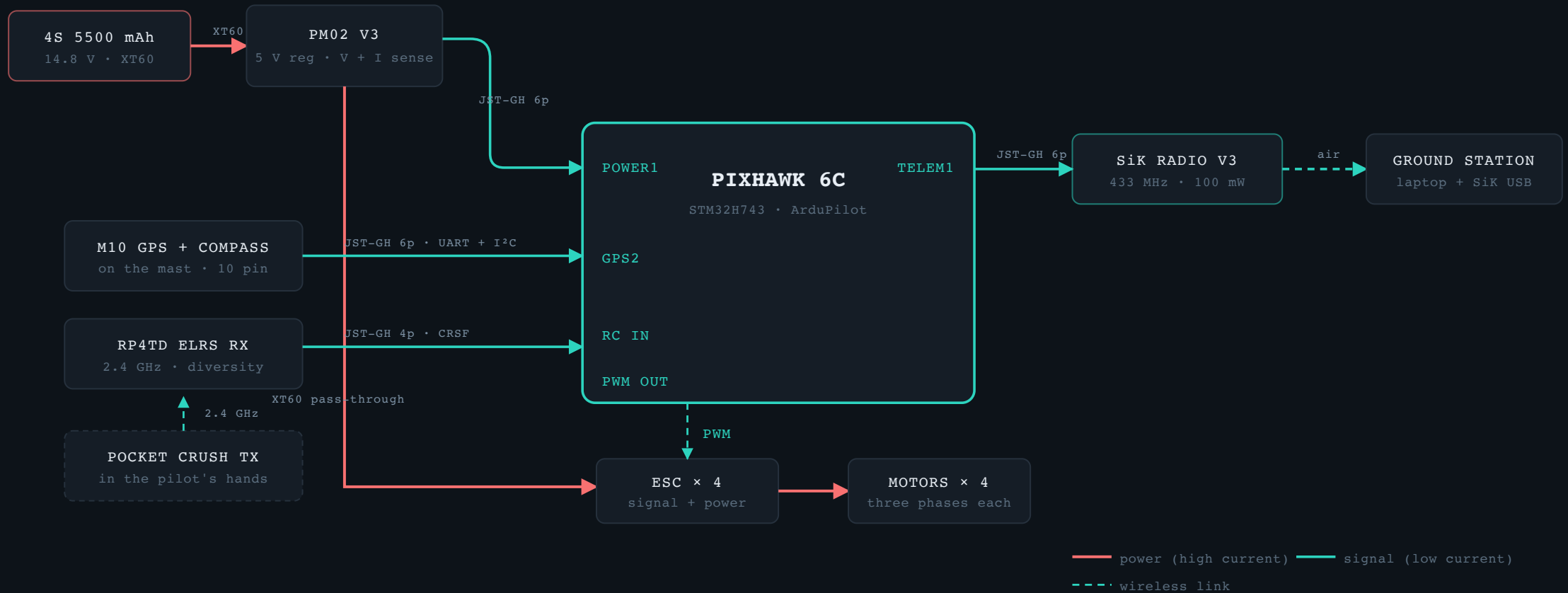
6-pin JST-GH into the Pixhawk. TELEM1 has its own **1.5 A power budget** — separate from the shared rail that feeds everything else. The SiK radio draws about 100 mA at full transmit, so there is plenty of margin.

433 MHz — WHY NOT 915?

The 915 MHz band is **not licence-free in Saudi Arabia**. 433 MHz is. The radios ship as a matched pair on the same frequency — do not mix them with a 915 MHz unit from another kit.

Plug the air unit into TELEM1, mount it with the antenna pointing up and clear of the frame. **When we open Mission Planner tomorrow, this is the cable the data travels down.**

The complete aircraft — every cable



Five signal cables into the Pixhawk, one power cable from the battery, and the high-current path that bypasses the board entirely. **If you can trace every line on this diagram, you can wire the aircraft.**